

SHARANYA MISHRA

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Portfolio: <https://www.sharanya-mishra.com/>

EDUCATION

Illinois Institute of Technology, Chicago, IL	May 2026
Master of Science- Data Science - GPA: 3.7	
PES University, Bengaluru, India	June 2024
Bachelor in technology - Computer Science/Engineering-GPA: 3.5	

SKILLS

Programming & Tools: Python, R, SQL, C++, JavaScript, MATLAB, Git, Docker, Kubernetes
AI / ML & Data Science: Machine Learning, Deep Learning, NLP, Computer Vision, Regression, Time-Series Forecasting (ARIMA, SARIMA, GARCH), Feature Engineering, Model Evaluation
Data Analysis & Statistics: Probability, Hypothesis Testing, Statistical Inference, Process Optimization, Work Measurement, A/B Testing
Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Excel (Pivot Tables, VLOOKUP, Variance Analysis), SAP
Cloud & Big Data: AWS (S3, EC2, SageMaker), Microsoft Azure, PySpark, Hadoop, Jenkins

EXPERIENCE

Lead Research Assistant January 2025 - Present

Illinois Institute of Technology

- Led a 4-member team in developing **OptiWeb**, a browser-based trajectory optimization platform for robotics applications.
- Automated optimization model generation and benchmarked performance across **50+** NEOS solver configurations.
- Improved simulation performance by **~30%** and integrated LLM-driven assistance for optimization setup and analysis.

Data Science Consultant / Project Lead

January 2026 – May 2026

LabelMaster Practicum

- Analyzed multi-year customer, transaction, product, and location datasets (2012–2026) to study **churn, retention, and revenue concentration**.
- Identified a key trend: active customers declined while revenue per customer increased, highlighting **account concentration risk**.
- Built **customer segmentation tiers** using revenue contribution, purchase activity, geography, and dormant-account behavior.
- Found win-back opportunities among inactive but historically valuable customers across priority states.
- Presented stakeholder-ready recommendations for retention campaigns, geographic targeting, and customer reactivation strategy.

Data Classification- Spring Intern

January 2024 - July 2024

Aptos Retail

- Built **15+ Azure-based data validation** rules to classify sensitive enterprise data and improve reporting accuracy to **95%**.
- Automated classification workflows across **5+** teams using Microsoft Purview and Salesforce, reducing manual effort by **40** .
- Supported **data governance** by improving data visibility, ownership tracking, and compliance-ready reporting.

Data Analyst Summer Intern (Risk Advisory)

June 2023 - August 2023

Deloitte Touche Tohmatsu LLC

- Automated** anomaly detection across **7+ client datasets** using **Python** and **SQL** to identify reporting inconsistencies.
- Built Excel dashboards using **Pivot Tables**, **VLOOKUP**, and variance analysis to support risk advisory decisions.
- Cleaned and validated client data to improve audit-readiness and reporting efficiency.

Data Analyst Intern

June 2022 - August 2022

Capgemini

- Developed **SQL-based analytics** pipelines and **Tableau dashboards**, enhancing client reporting efficiency.

RESEARCH PUBLICATION

Audio Description for the Visually Impaired (IEEE, 2024)

Link to paper: <https://ieeexplore.ieee.org/document/10544216>

GitHub: https://github.com/Obsarian/Audio-Description-of-Videos_Capstone

Technologies: Python, PyTorch, YOLO, ResNet-35 (Kinetics), Google Text-to-Speech API

- Developed a **deep learning pipeline** to generate **real-time voice captions** from videos, achieving **BLEU score 0.7**.
- Combined **object** and **action recognition** models to enhance **video accessibility** for visually impaired users.

PROJECT

InsightAI - Production RAG Agent for Google Workspace

GitHub <https://github.com/mishrasharanya/Insight--AI-agent> | **Live Demo:** <https://insight-ai-agent-phi.vercel.app/>

Technologies: React, FastAPI, Supabase (pgvector), Groq/Llama, OAuth 2.0, Python, JavaScript

- Built a **4-agent RAG system** for Google Workspace with semantic retrieval and grounded responses.
- Engineered Planner, Retrieval, Reflection, and Evaluation agents with confidence scoring and PII redaction.
- Deployed on **Vercel, Render, and Supabase** using OAuth 2.0 and pgvector.

HerWay - AI-Powered Urban Analytics Platform

GitHub: github.com/mishrasharanya/HerWay-Soremo | **Live Demo:** <https://her-way-soremo.vercel.app/>

Technologies: Python, NLP, OpenAI API, React, Leaflet

- Built a multi-source urban analytics platform combining Reddit, 150k+ 311 service requests, and crime datasets to generate neighborhood-level insights across **77 Chicago community areas**
- Developed one of the only fellowship projects integrating **NLP** and **OpenAI-powered analysis** to extract community signals from 1,000+ public posts and records.
- Built an AI-powered chatbot and geospatial dashboard for community-level insights and decision making.

ForeQuest - AI-Powered Financial Forecasting Platform

GitHub: github.com/mishrasharanya/Forequest-Forecast_Smarter | **Live:** <https://forequest.streamlit.app/>

Technologies: Python, Streamlit, GARCH, Monte Carlo, Groq, LLMs

- Built an end-to-end financial forecasting platform integrating ARIMA, GARCH, Monte Carlo simulation, option pricing models, and 10+ years of market data for stock trend and volatility analysis.
- Achieved **~85% R²** and **<0.02 MAE** in volatility prediction while integrating **Groq-hosted LLMs** to deliver explainable forecast interpretation, risk analysis, and conversational financial insights through a deployed **Streamlit** application.